

FINAL YEAR PROJECT • WSC 2026 SEE PAKISTAN

# SMART MINUTES

AI-powered meeting recorder & summarizer

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8.3%

Word error rate

4.2/5

AI summary quality

1.4s

Caption latency

3

Role tiers



# What we'll cover

**01** Introduction

**02** Problem statement

**03** Solution & objectives

**04** Scope

**05** Literature review

**06** Tools & technologies

**07** System architecture

**08** Requirements

**09** UML diagrams

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# What is Smart Minutes?

An AI-powered desktop application that automates the entire meeting documentation lifecycle — from live recording to structured, distributed minutes.



## Live & post-meeting STT

Real-time captions via Vosk; high-accuracy transcripts via OpenAI Whisper



## AI summarization

Google Gemini 1.5 Pro generates structured minutes with decisions and topics



## Offline + online

Supports both in-person and remote meetings in one platform



## Three-tier RBAC

Super Admin, Admin, and User roles for organizational governance



## Modern stack

Flutter desktop frontend with a FastAPI / Python backend



## Incremental delivery

Built using the Incremental Life Cycle Model across 5 stages

# Why meetings need a smarter system

✗ Manual note-taking is time-consuming and distracts participants

✗ Decisions and action items are frequently missed or forgotten

✗ Existing tools (Otter.ai, Fireflies.ai, Teams) are cloud-only

✗ No tool offers full offline support for in-person meetings

✗ Current solutions lack role-based access and org management

✗ No system unifies task extraction, tracking, and PDF minutes

# The Smart Minutes solution

- ✓ Real-time + post-meeting transcription (Vosk + Whisper)
- ✓ Structured minutes generation via Google Gemini 1.5 Pro
- ✓ Three-tier role-based access control
- ✓ Automatic task extraction, assignment & notifications
- ✓ Semantic analysis: tone, topic relevance, keywords
- ✓ Hybrid offline/online support with secure storage
- ✓ Professional PDF reports with one-click email sharing

# Scope



## Target users

Corporate, academic, and government organizations



## Platforms

Windows 10/11, macOS 12+, Ubuntu 20.04+ (desktop only)



## Language

English only — Vosk and Whisper configured for English



## Speaker ID

Based on text context, not audio characteristics



## Accuracy limits

May drop in high-background-noise environments



## Internet dependency

AI summarization requires internet (Gemini API)

# Literature review

- |                          |                                                           |
|--------------------------|-----------------------------------------------------------|
| ● Vosk                   | Offline, Kaldi-based ASR with multilingual support        |
| ● OpenAI Whisper         | Trained on 680k hours of audio; robust to accents & noise |
| ● Google Gemini 1.5 Pro  | Long-context NLP for summarization & extraction           |
| ● Otter.ai               | Online only — no role management or formal minutes        |
| ● Fireflies.ai           | Subscription-based — no offline support                   |
| ● Google Meet / MS Teams | Cloud-locked — no PDF minutes generation                  |

Smart Minutes is the only solution offering hybrid offline+online support, RBAC, and full org management

# Tools & technologies



## Frontend

Flutter 3.24+ with Dart — cross-platform desktop UI



## Backend

Python 3.9+, FastAPI, Uvicorn ASGI server



## AI / ML

Vosk, OpenAI Whisper, Google Gemini 1.5 Pro API



## Data

MySQL 8.0, SQLAlchemy ORM, Firebase Auth, JWT



## Real-time

WebSocket & WebRTC for live audio streaming

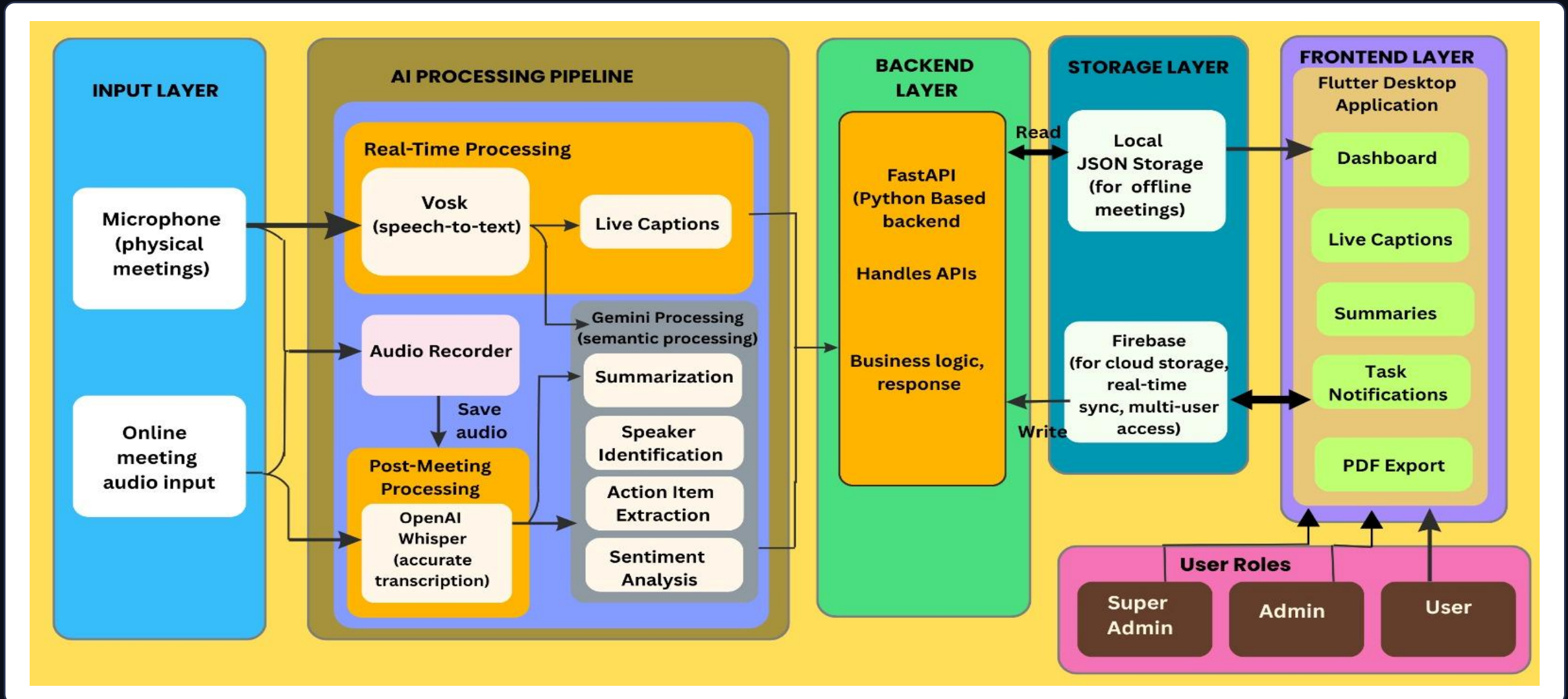


## Processing

FFmpeg for audio processing & WAV merging



# System architecture



# System modules



## Authentication

Login, registration, role assignment, sessions



## Meeting management

Create, view, edit, delete, start meetings



## Audio recording

Start/stop, WAV merging, disconnect handling



## Live transcription

Real-time captions, full transcript export



## AI summarization

Generate, edit, approve/reject (Admin only)



## Task management

Auto-extract, assign, track, notify



## User & dept mgmt

Role assignment, department creation



## Reports & analytics

PDF generation, email, dashboard stats

# Functional & non-functional requirements



## Functional

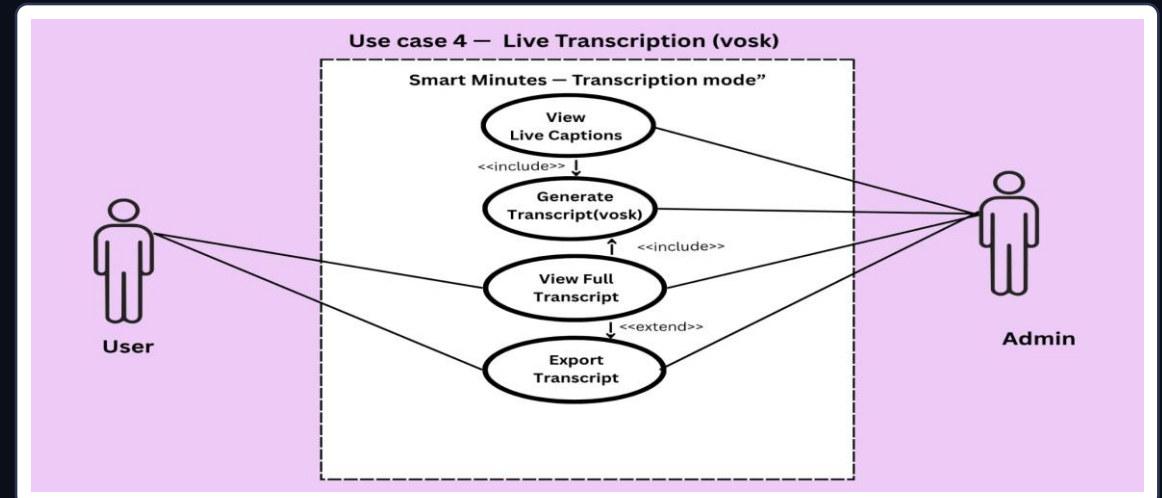
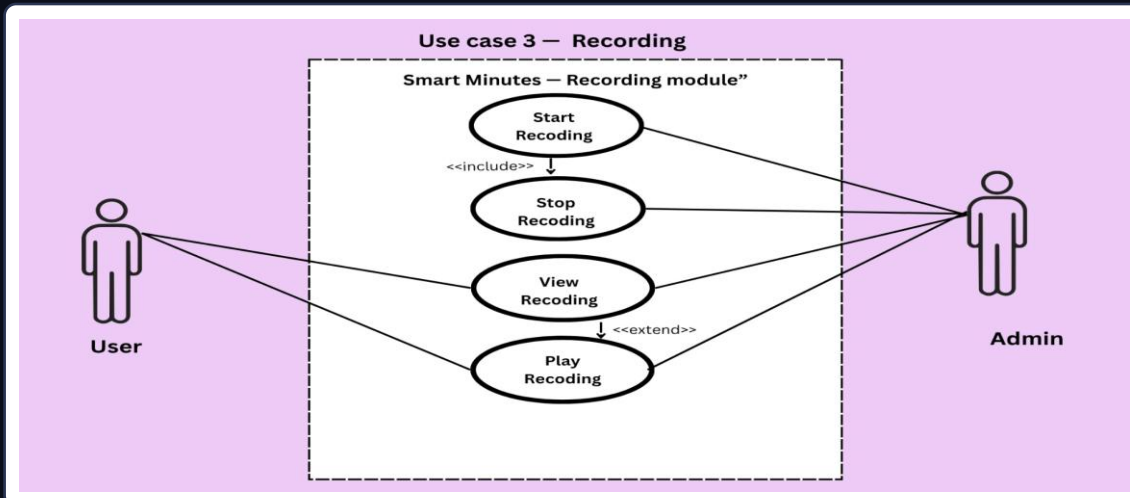
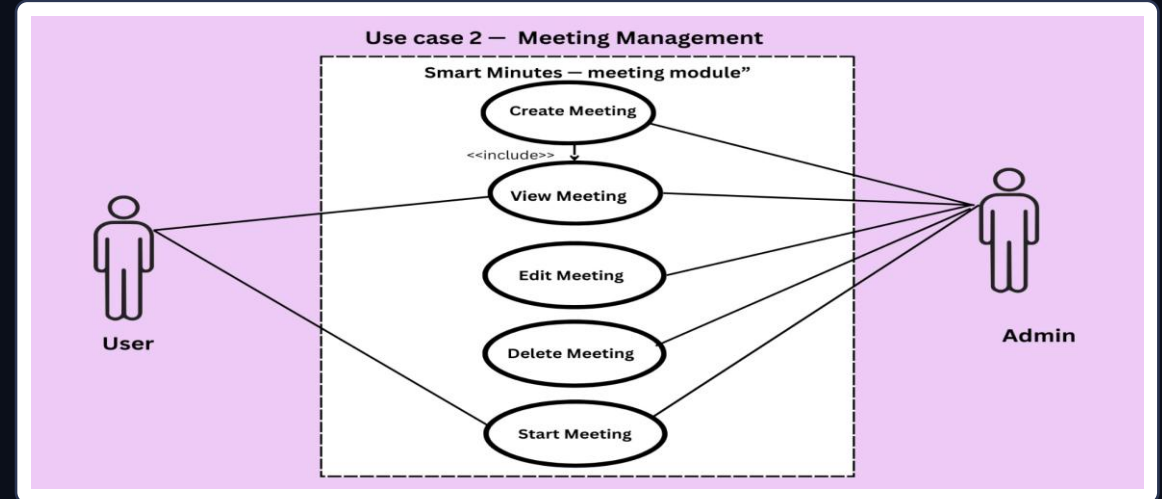
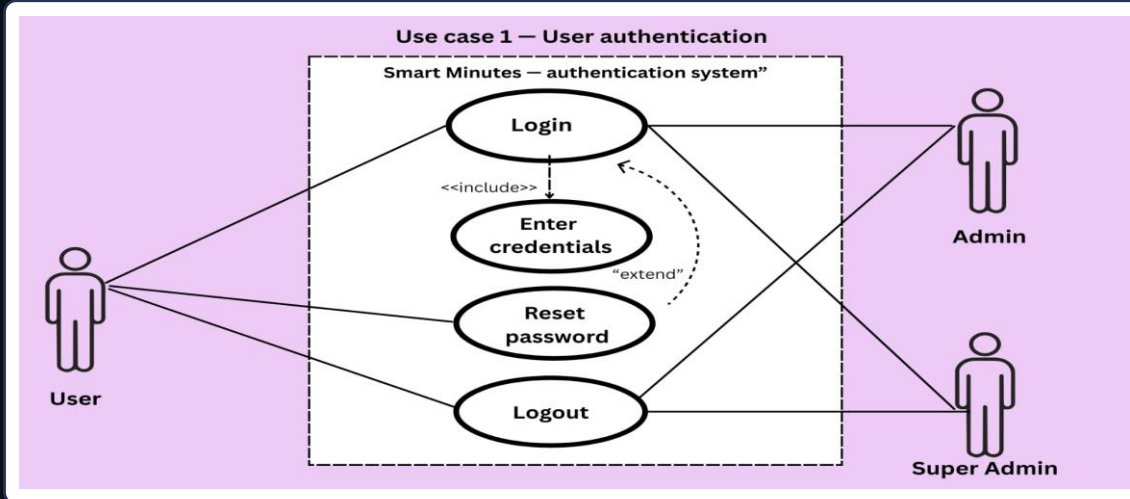
- ✦ Schedule, organize & track meetings with agenda
- ✦ Role-based secure access (Super Admin, Admin, User)
- ✦ Live capture with real-time captions & ASR transcripts
- ✦ Auto-generate summaries with decisions & action items
- ✦ Admin review, edit, and approval of AI-generated minutes
- ✦ Auto task creation, assignment & PDF + email distribution



## Non-functional

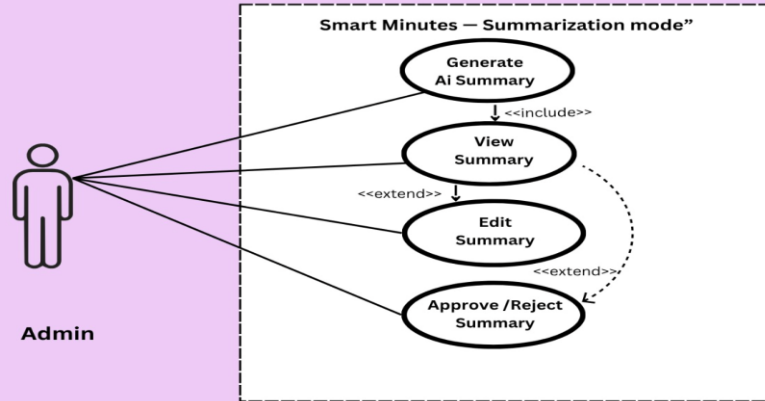
- ✦ Word Error Rate under 15%
- ✦ Live captions delivered within 2 seconds
- ✦ Cross-platform: Windows, macOS, Linux
- ✦ Secure: Firebase + JWT + RBAC enforced
- ✦ Usable by both technical and non-technical users

# Use case diagrams (1 – 4)

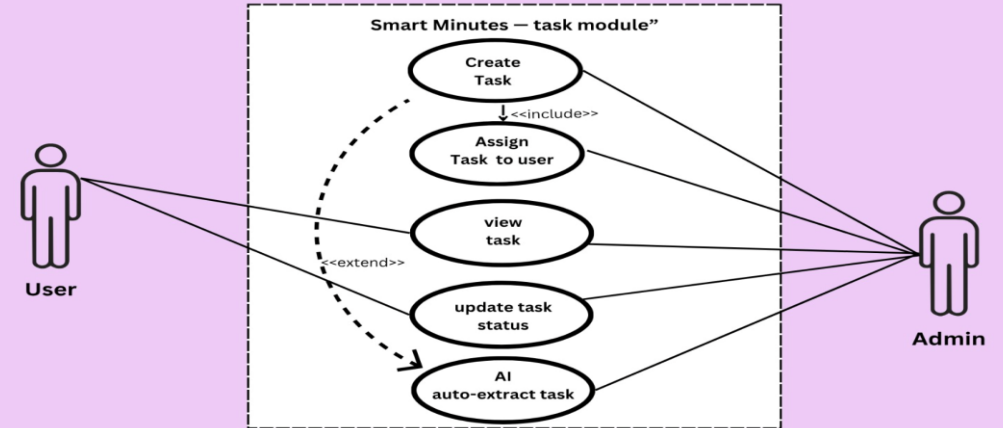


# Use case diagrams (5 – 8)

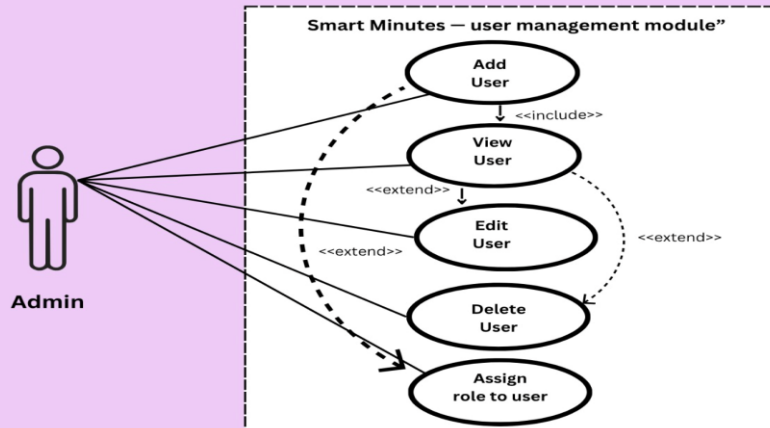
Use case 5 — Ai Summarization (GEMINI)



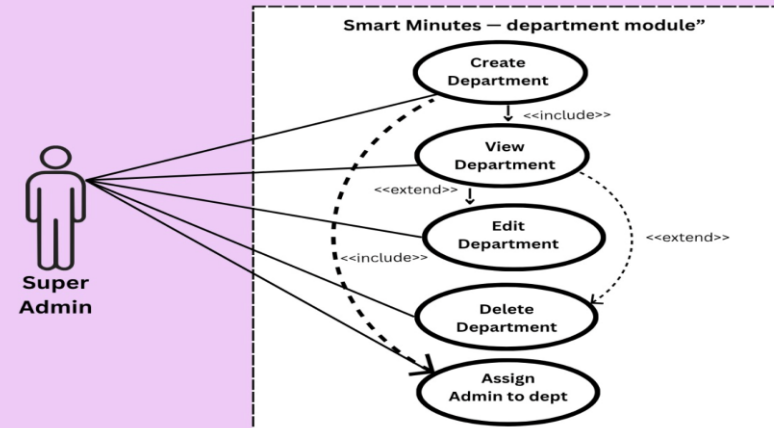
Use case 6 — Task Management



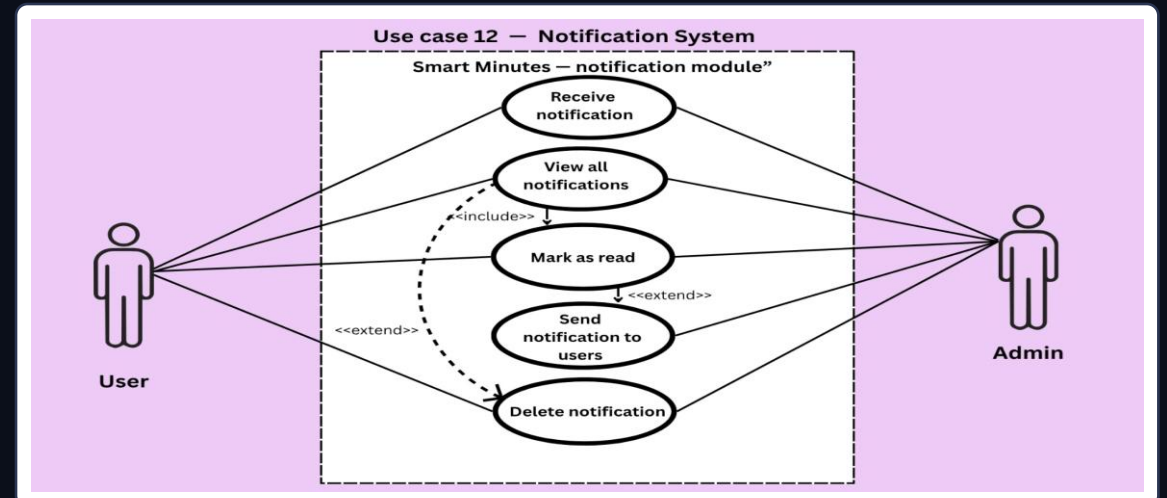
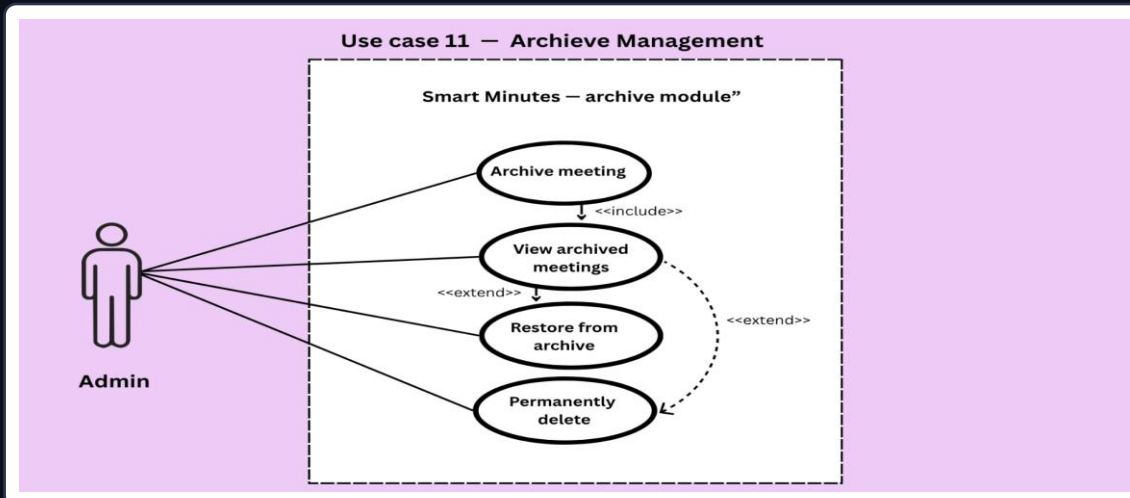
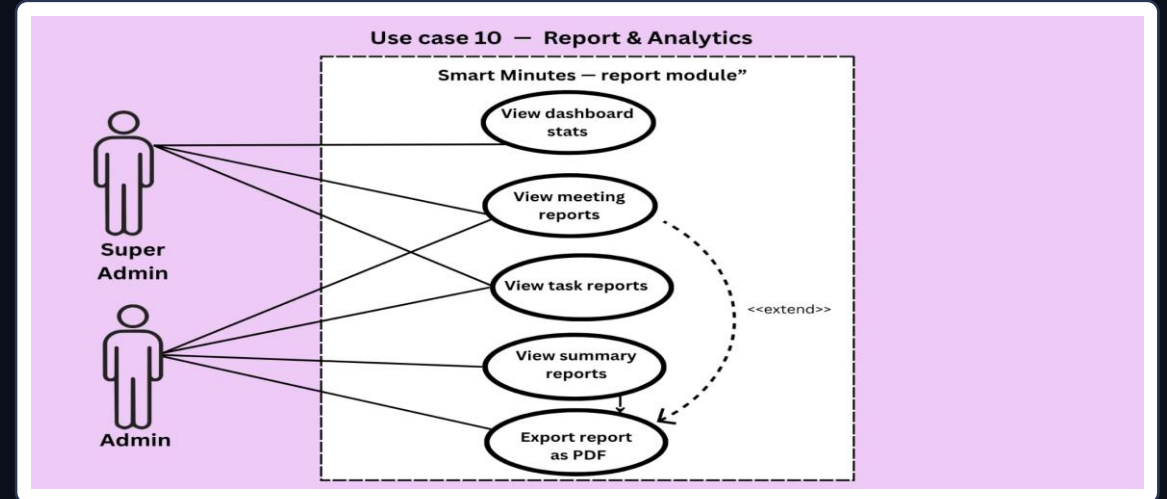
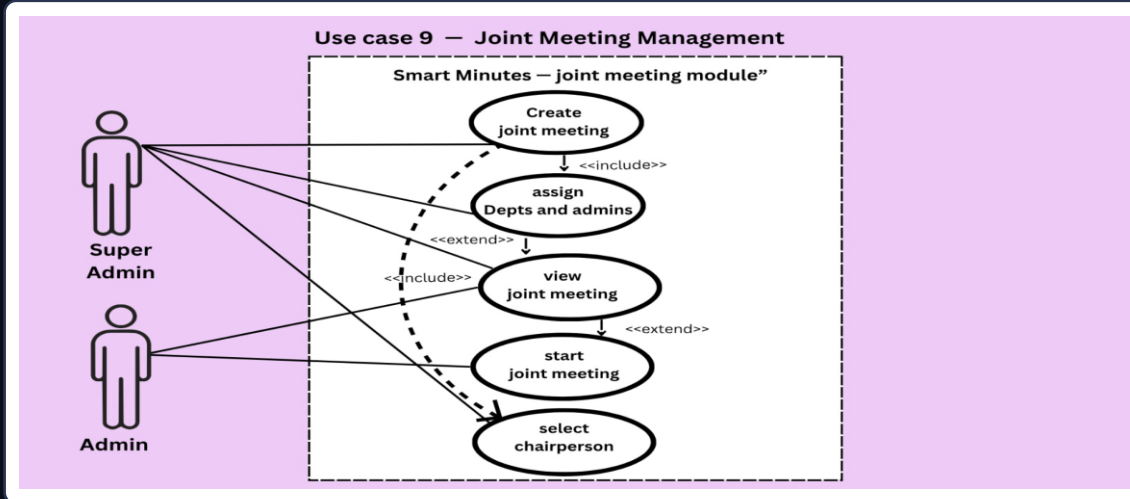
Use case 7 — User Management



Use case 8 — Department Management

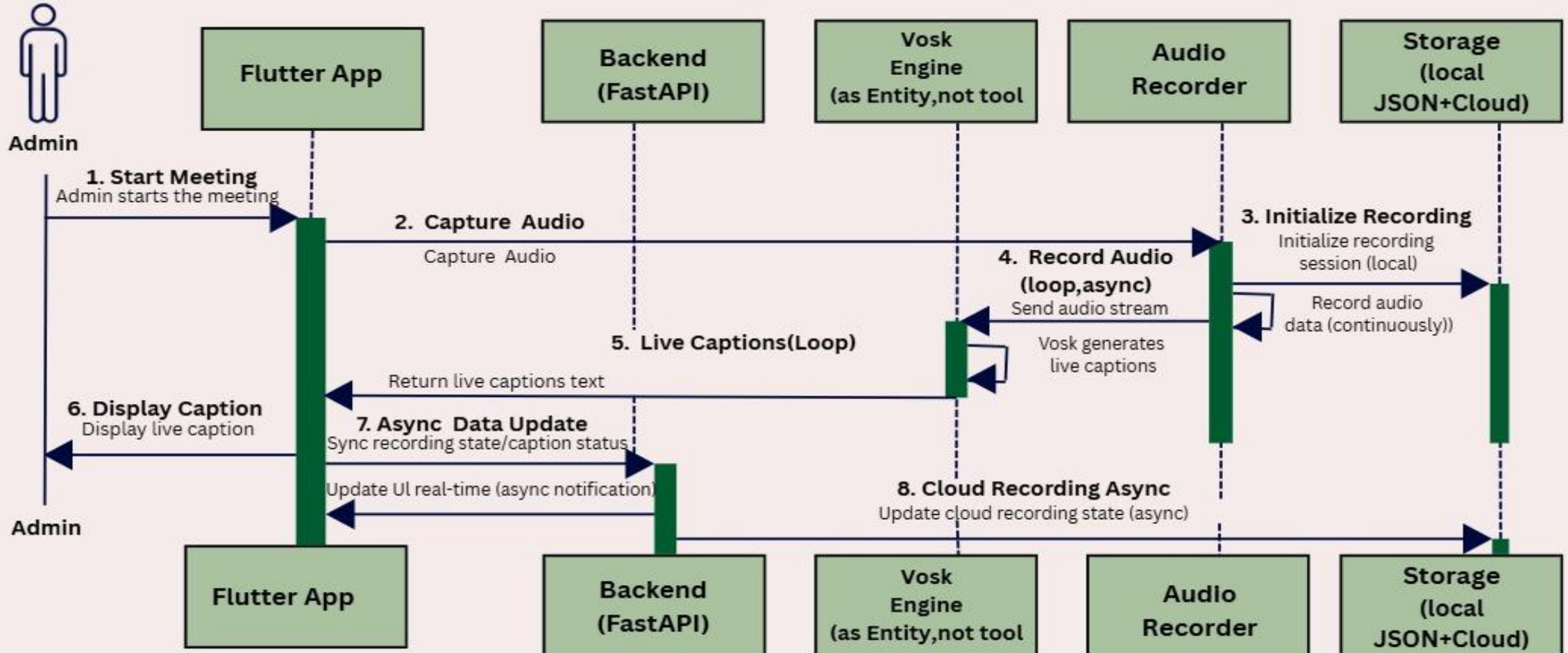


# Use case diagrams (9 – 12)

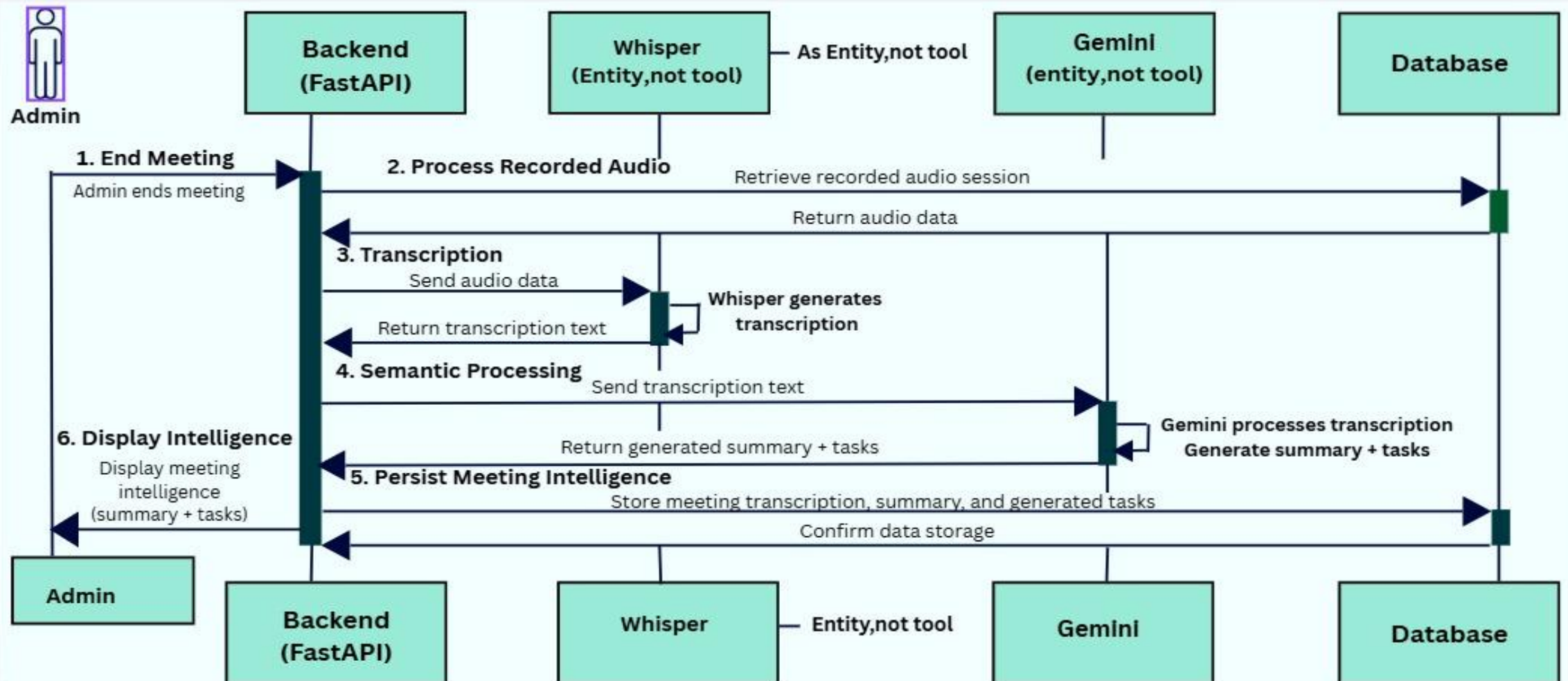




# Sequence diagram — live recording

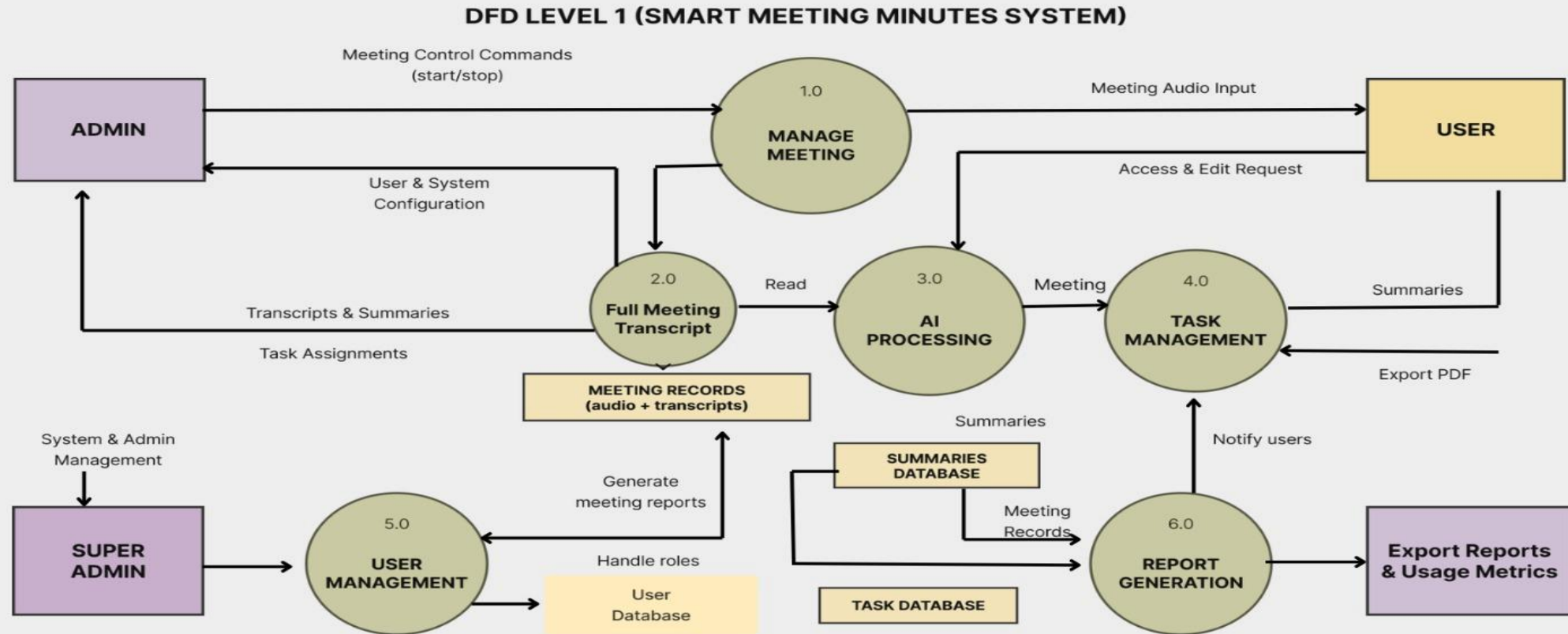


# Sequence diagram — AI processing

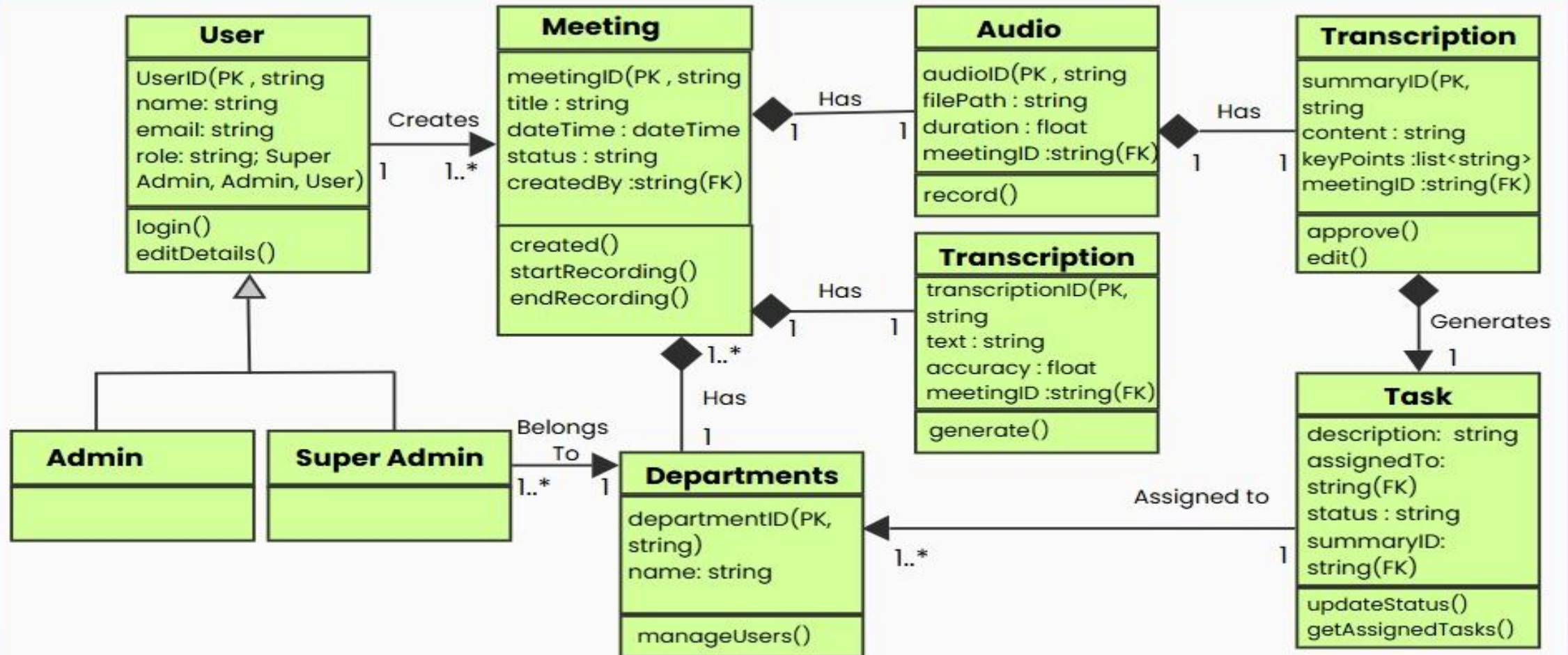




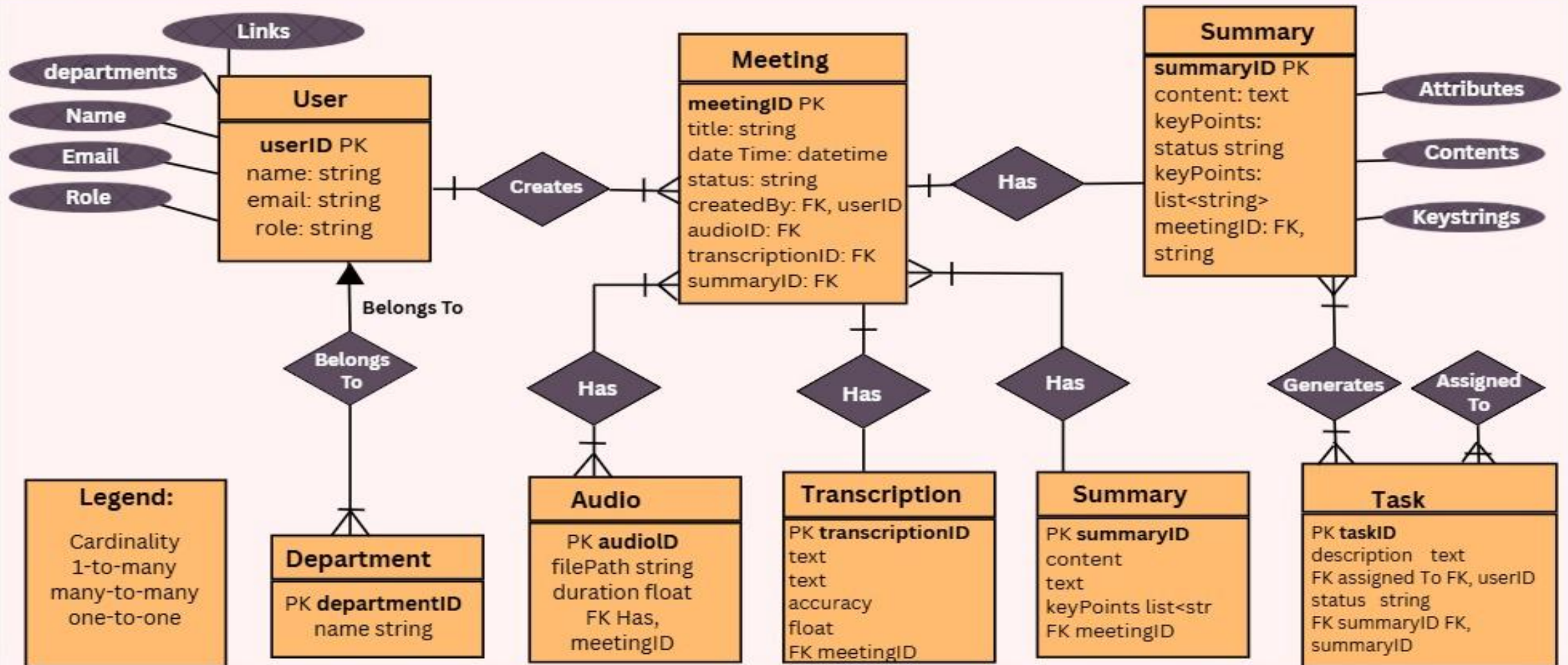
# Data flow diagram — level 1



# Class diagram

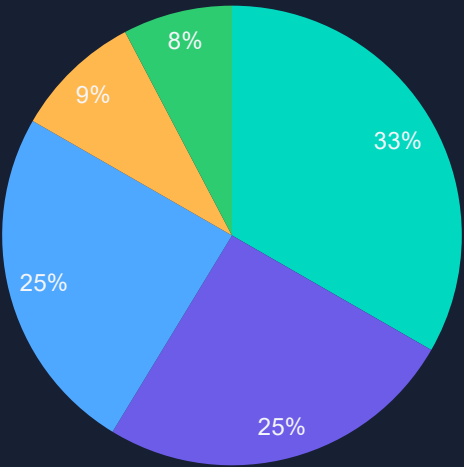


# Entity relationship diagram

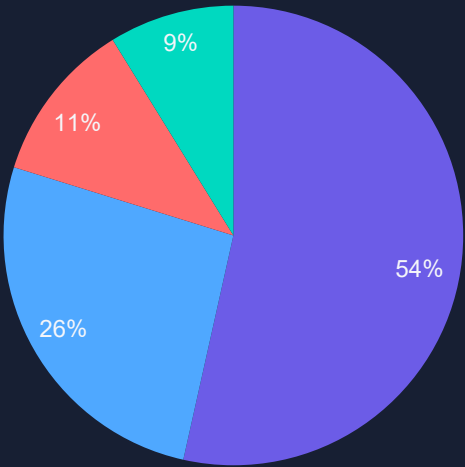


# Survey results — 114 responses

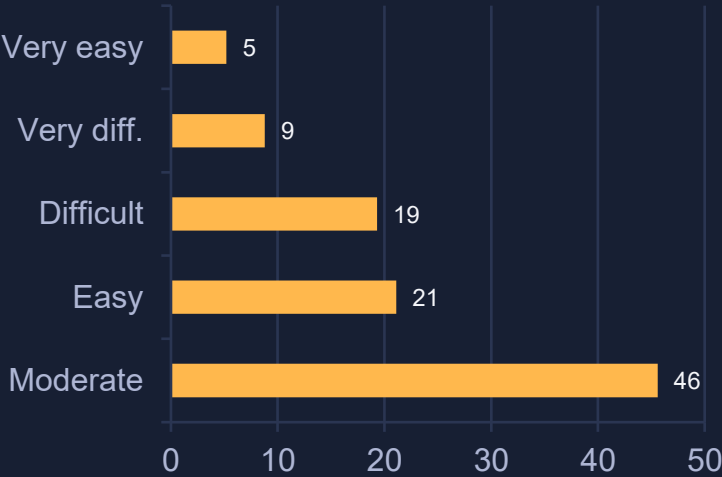
Sector breakdown



Meetings per month



Difficulty tracking action items



57%

Find auto-minutes  
extremely valuable

56.1%

Definitely/probably  
will use Smart Minutes

64%

Say it would  
genuinely help them

69.3%

Have never used  
an AI meeting tool

# Key survey insights



## Note-taking method

52.6% manually write notes • 23.7% typed during meetings • 13.2% audio recorded • 7% no formal method



## Time wasted per meeting

41.2% waste 20–30 min • 37.7% waste 10–20 min • 19.3% waste 30+ minutes on manual documentation



## Adoption intent

56.1% definitely/probably will use Smart Minutes • Only 1.8% said they definitely won't



## Would it help?

64% said YES it would genuinely help their org • 28.9% said maybe — very high positive signal



## AI tool experience

69.3% have never used an AI meeting tool — huge untapped market • Key issues: internet, accuracy, language



## Top requested features

Multi-language (Urdu) support • Calendar integration • Speaker identification • Reminder system

# User interface

Built with Flutter 3.24+ — a single cross-platform desktop codebase for Windows, macOS, and Linux.



## Login screen

Firebase-integrated, role-based secure access



## Dashboard

Role-specific layout for each user type



## Meeting screen

Record button, live captions, status indicator



## Summary review

Admin can view, edit, approve or reject AI summaries



## Task manager

View tasks, update status, get notifications



## Reports

PDF preview with one-click email distribution

# Login screen

sm\_frontend

## Welcome Back

Sign in to continue to Smart Minutes

Email \*

22112021007it@s.rwu.edu.pk

Password \*

☒ Remember Me

[Forgot Password?](#)

Login

Demo Credentials:

Use credentials sent by email after account provisioning.

### Quick Login

humaminhas14314@gmail.com

SUPER\_ADMIN

×

➤

22112021007it@s.rwu.edu.pk

ADMIN

×

➤

aansahfayyaz13388@gmail.com

USER

×

➤

minhaszillehuma36@gmail.com

ADMIN

×

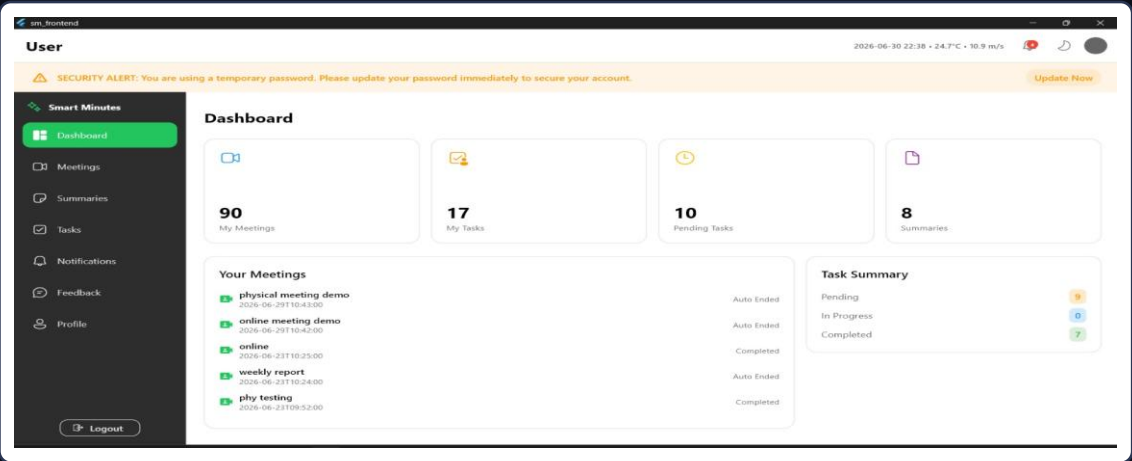
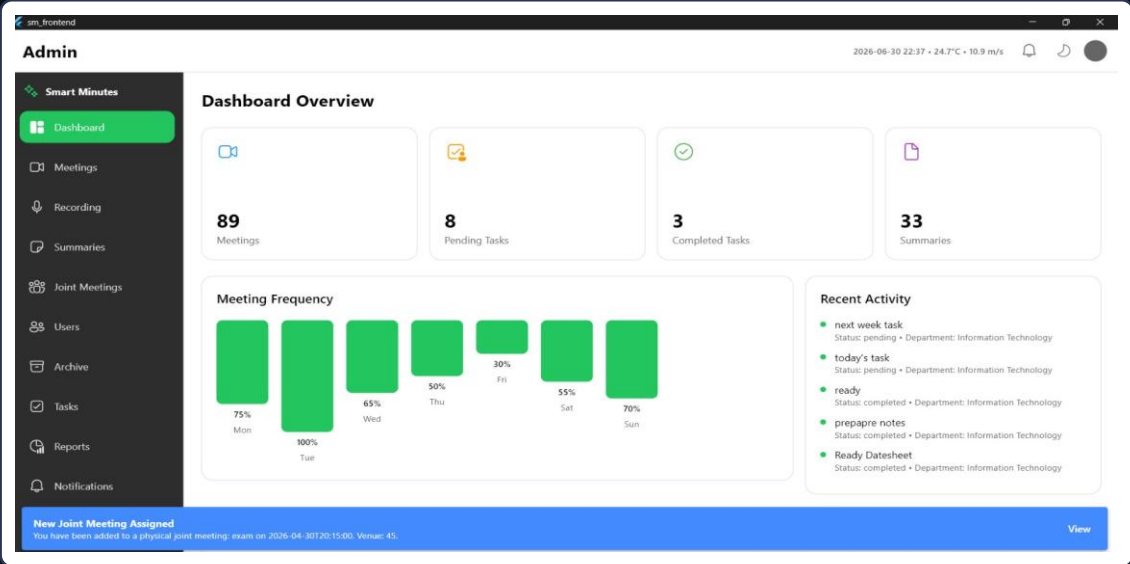
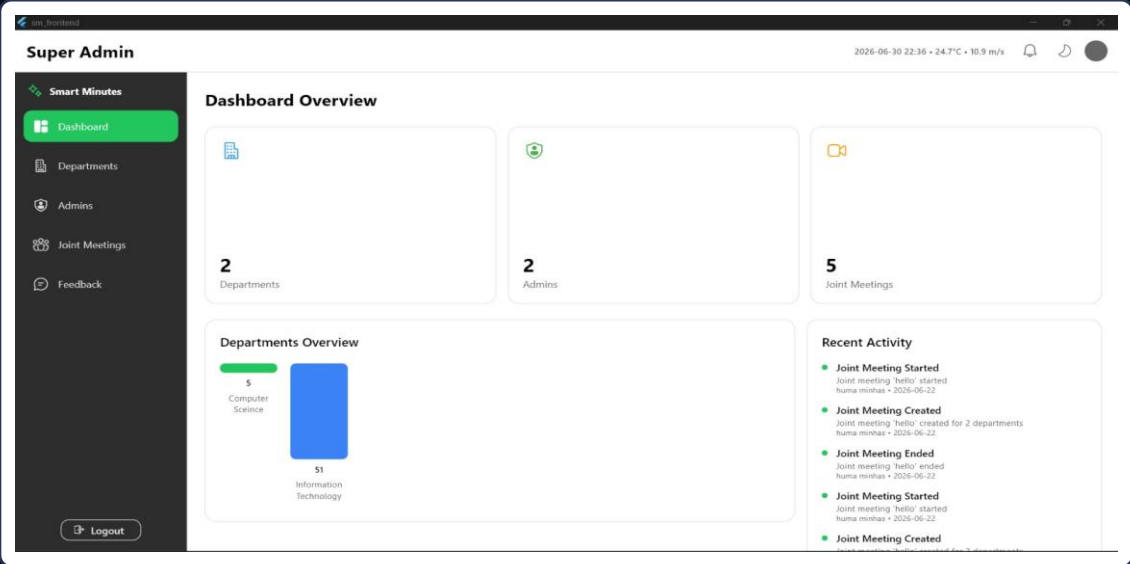
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SMART MINUTES

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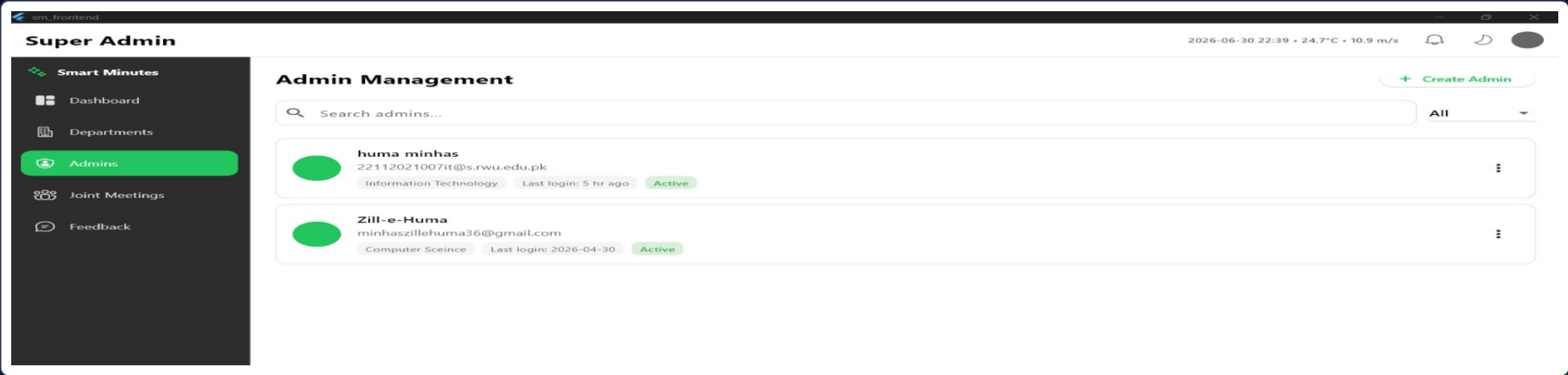
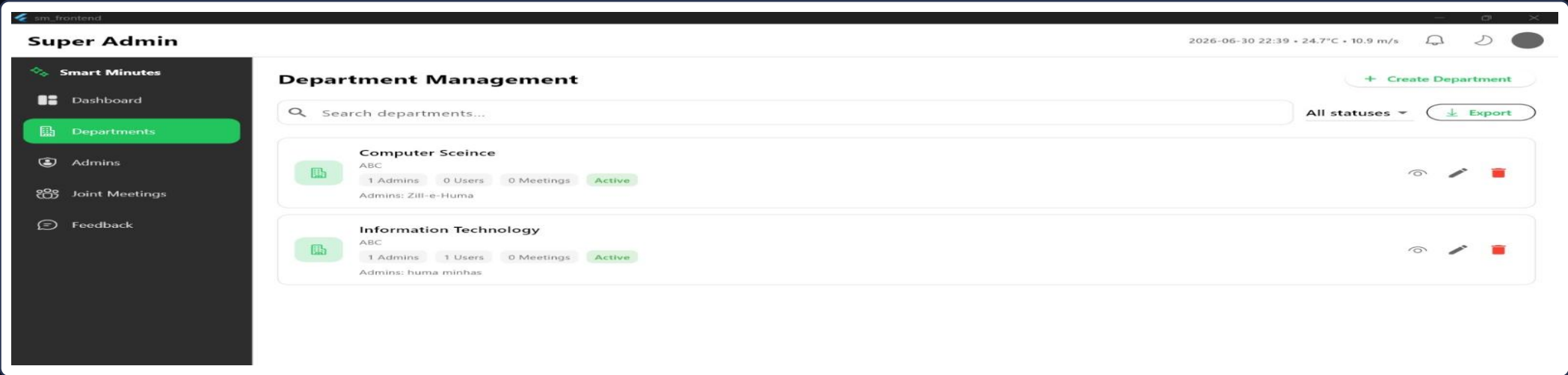


# Role-based dashboards

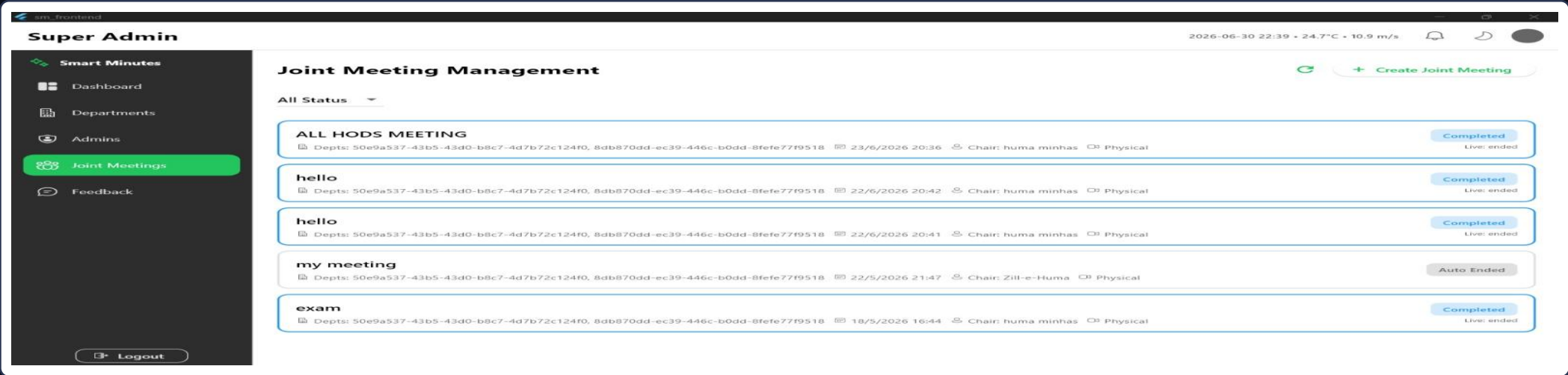




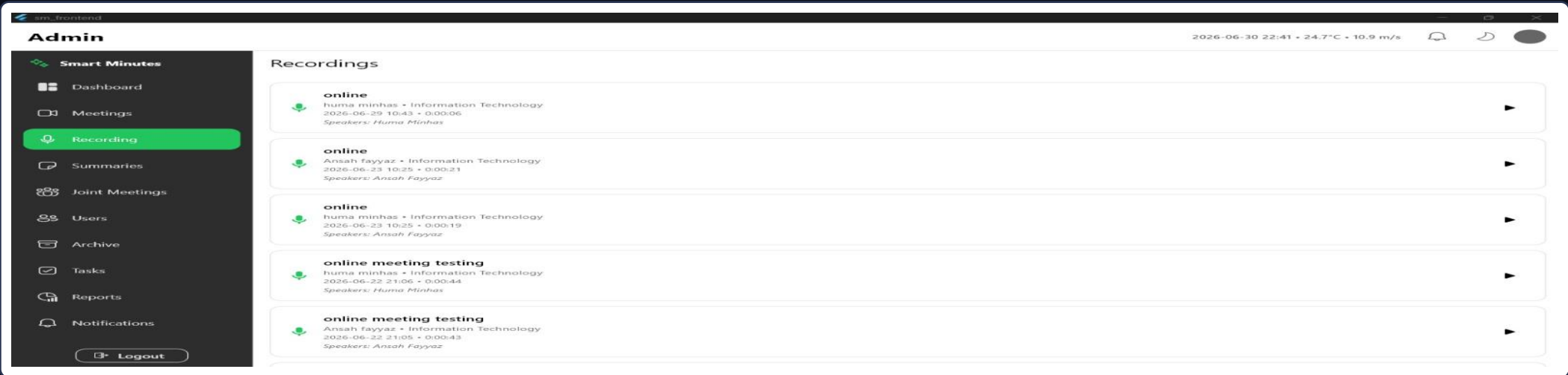
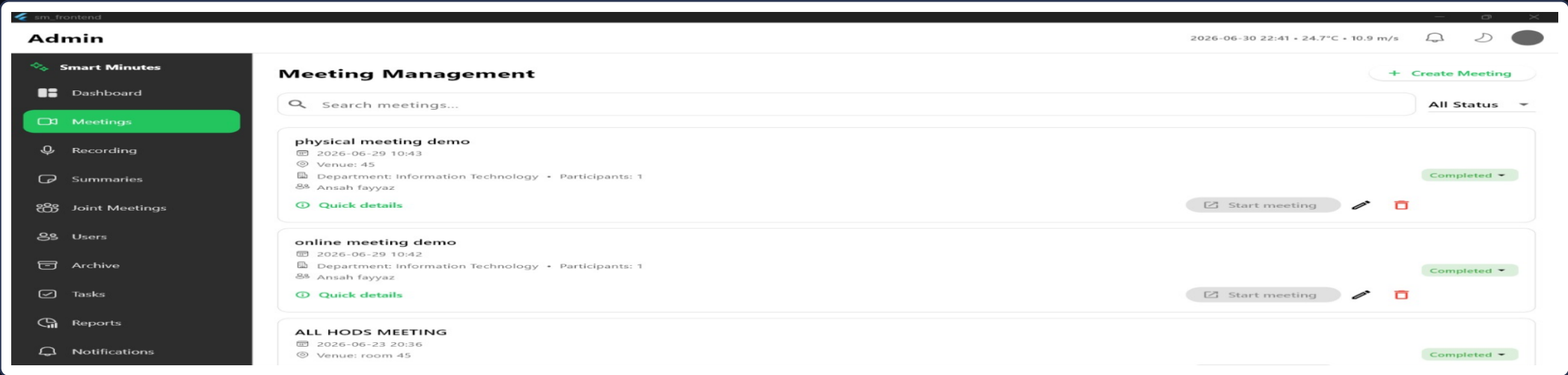
# Department & admin management



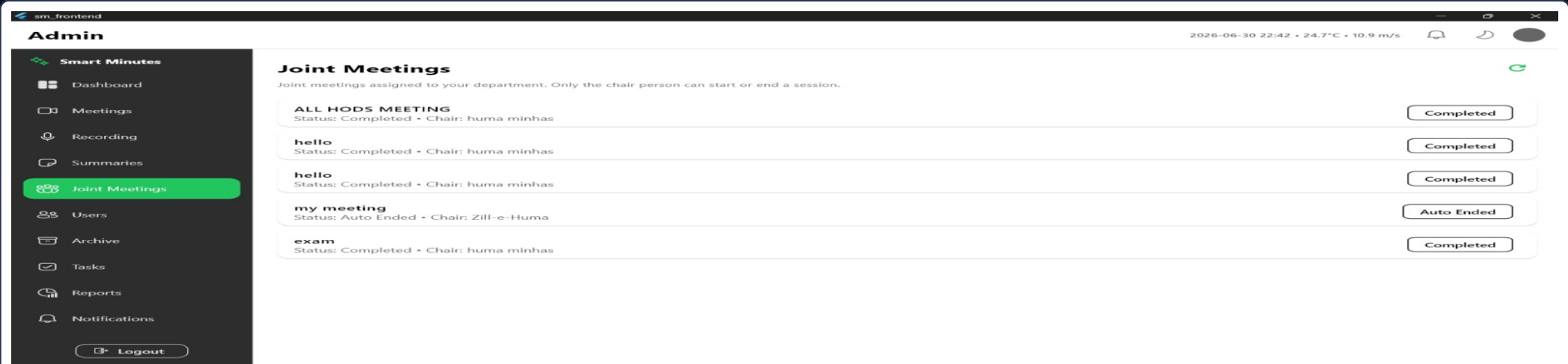
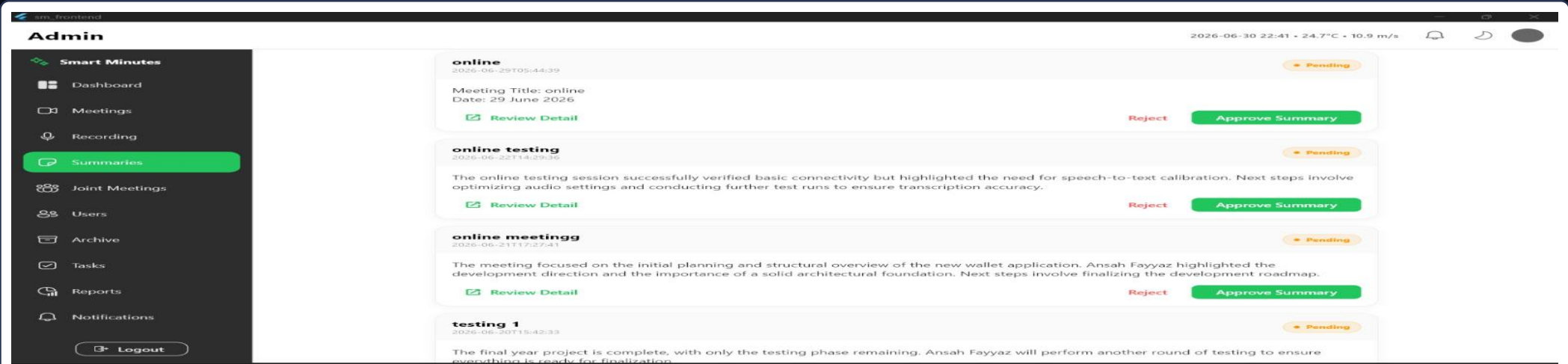
# Joint meetings & feedback



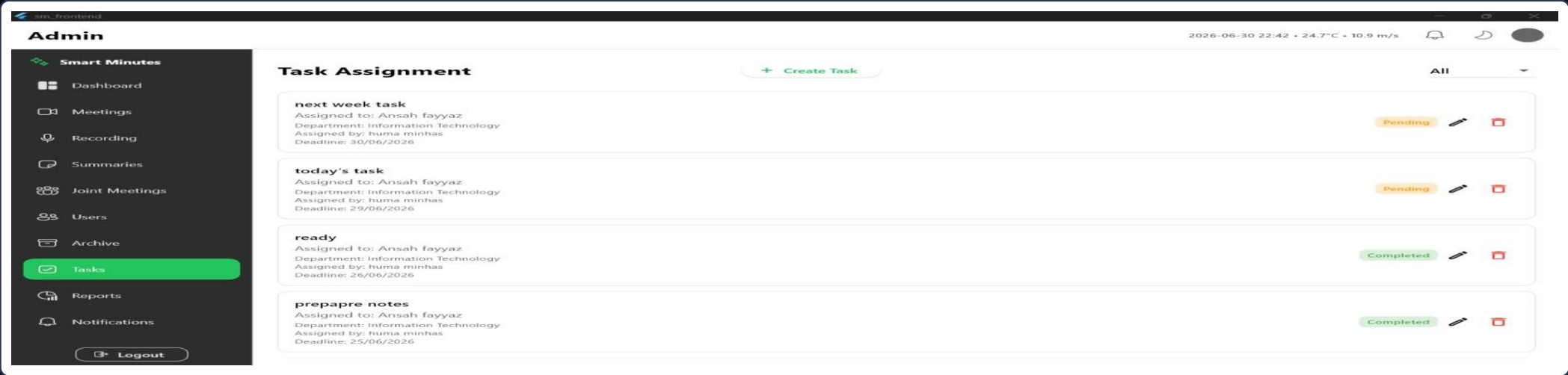
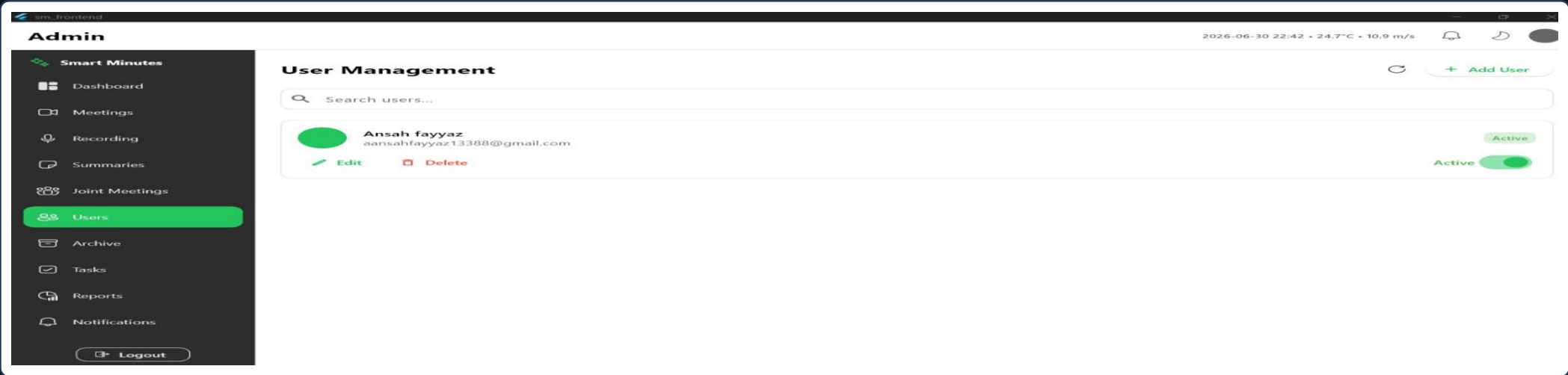
# Meeting & recording views



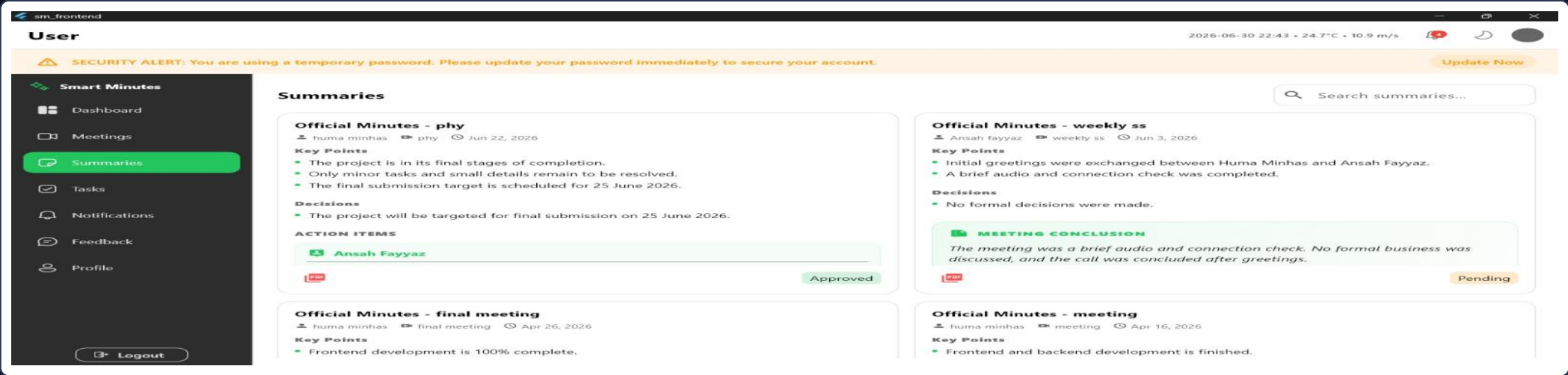
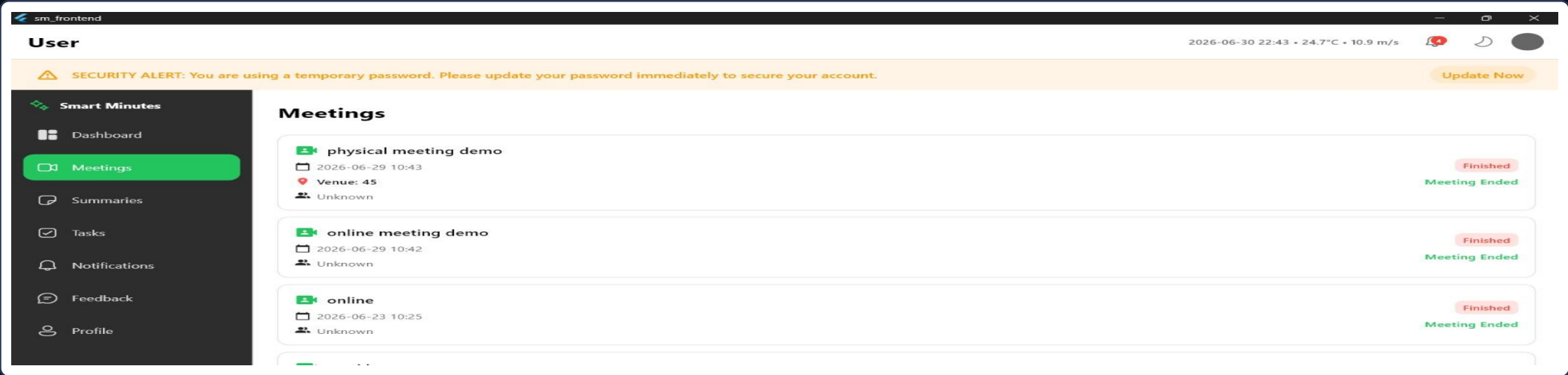
# Summary approval workflow



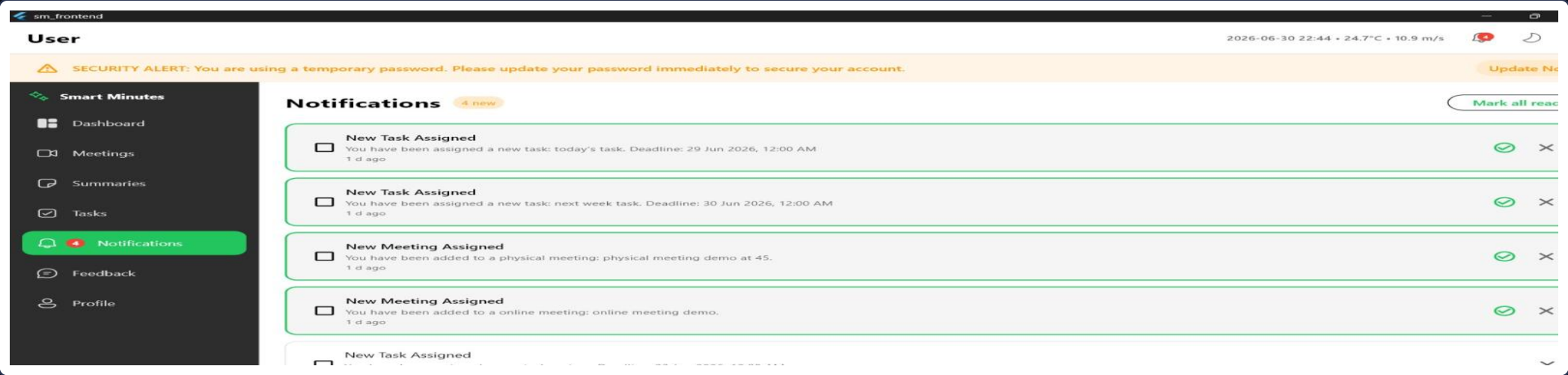
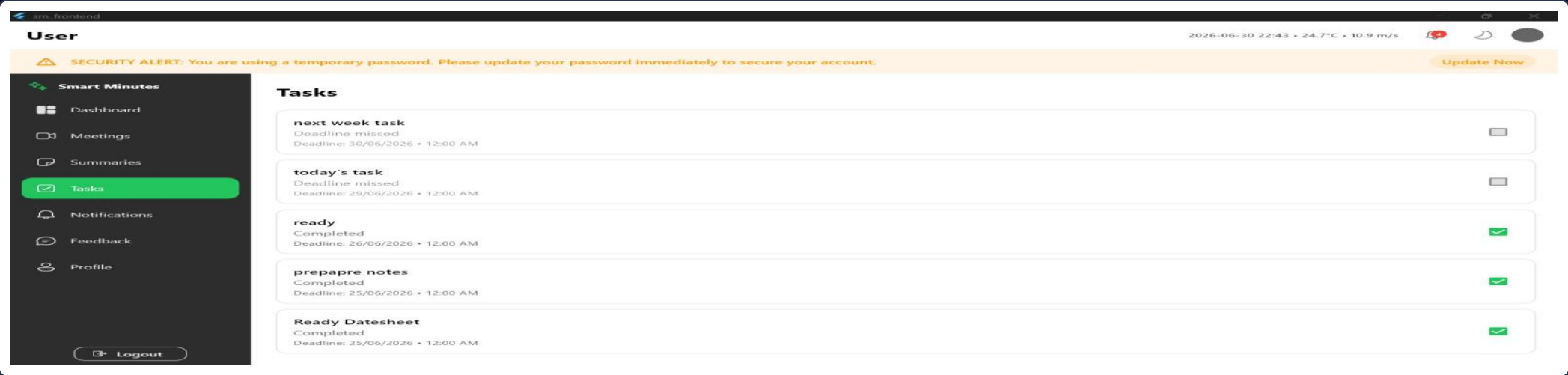
# User & task management



# User meeting views



# Tasks & notifications

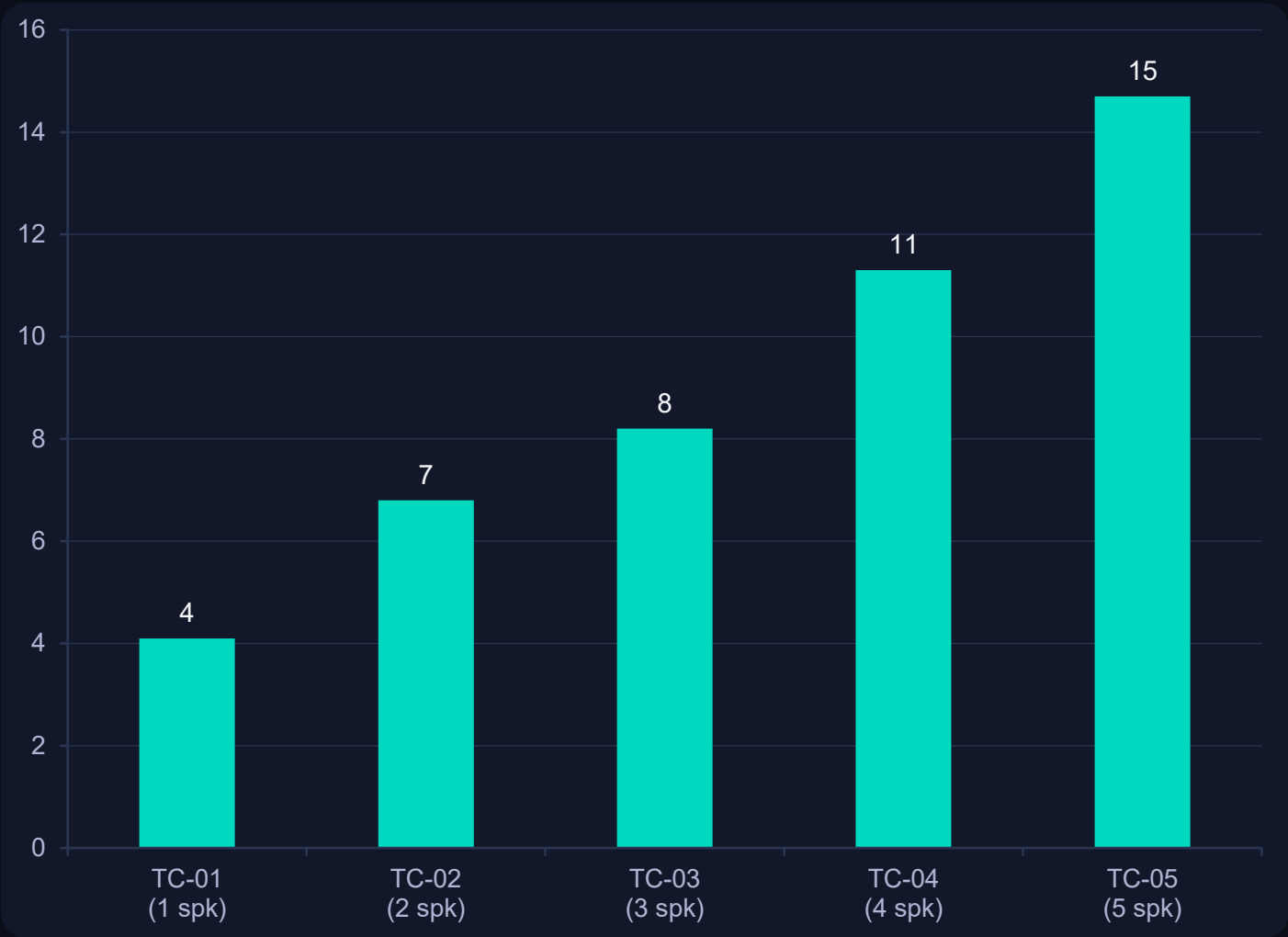


# Functional test cases

| ID    | Module    | Test case                    | Status |
|-------|-----------|------------------------------|--------|
| TC-01 | Auth      | Login with valid credentials | Pass   |
| TC-02 | Auth      | Login with invalid password  | Pass   |
| TC-03 | Recording | Start meeting recording      | Pass   |
| TC-04 | Recording | Stop recording and save WAV  | Pass   |
| TC-05 | STT       | Transcribe 10-min audio      | Pass   |
| TC-06 | AI        | Generate meeting summary     | Pass   |
| TC-07 | Admin     | Approve meeting summary      | Pass   |
| TC-08 | PDF       | Export meeting minutes PDF   | Pass   |
| TC-09 | Email     | Send minutes via email       | Pass   |
| TC-10 | Online    | Record online meeting        | Pass   |



# Transcription accuracy results



Average WER

8.3%

Target was under 15% — goal exceeded

**Quiet office**

1 speaker, 10 min

**Conference room**

2 speakers, 15 min

**Meeting room**

3 speakers, 20 min

**Open office**

4 speakers, 30 min

**Background noise**

5 speakers, 45 min

# Smart Minutes drives real SDG impact

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## DECENT WORK & ECONOMIC GROWTH

- Automates documentation — frees professionals for high-value productive work
- Reduces meeting overhead costs for SMEs and enterprises
- Women-led tech innovation: 3 jobs created, replicable model for IT Pakistan

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## INDUSTRY, INNOVATION & INFRASTRUCTURE

- First Pakistani offline-first AI meeting intelligence platform
- Built on a production-grade AI stack: Vosk, Whisper, Gemini 1.5 Pro
- Proves emerging-market universities can build export-ready AI products

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## PEACE, JUSTICE & STRONG INSTITUTIONS

- Tamper-resistant minutes for government and academic institutions
- Role-based access ensures accountability and transparent decision trails
- Offline-first design enables air-gapped government environments

# Conclusion & future work



## Conclusion

- ✦ Complete AI-powered, privacy-centered meeting system
- ✦ WER 8.3% • Summary score 4.2/5 • Caption delay 1.4s
- ✦ All performance targets met
- ✦ Hybrid recording, 3-tier RBAC, automated tasks, and PDF distribution all delivered



## Future work

- ✦ Speaker diarization using pyannote.audio
- ✦ Multi-language support for Vosk and Whisper
- ✦ Cloud database migration and mobile app development
- ✦ Local LLM integration to reduce Gemini API dependency



# THANK YOU

Questions & discussion

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